Assignment-2

"Take one Domain and draw the graph (Normal distribution) (Empirical rule)"

Introduction:

Normal distribution, also called the **Gaussian distribution**, is a way to describe how values in a dataset are spread. In a normal distribution, most values are concentrated around the **mean (average)**, and fewer values appear as we move further away from the mean. This creates a **bell-shaped curve** when plotted on a graph.

It is one of the most common distributions in statistics and real life. Many natural and social phenomena, like **students' exam scores, heights, weights, and IQ scores**, follow a normal distribution.

Importance of Normal Distribution:

1. Helps understand the **pattern of data** and how values are spread.
2. Used to calculate **probabilities** and predict outcomes.
3. Forms the basis for many **statistical tests** and analyses.
4. Helps identify **outliers** or unusual values in data.
5. Makes it easier to apply the **empirical rule (68-95-99.7%)** to estimate percentages of data within certain ranges.

Domain Selection: Students' Exam Scores

For this report, we have chosen **students' exam scores** as the domain to demonstrate normal distribution and the empirical rule. Exam scores are one of the most common examples in statistics that naturally follow a **normal distribution**, making it easy to understand patterns, averages, and variations in real life.

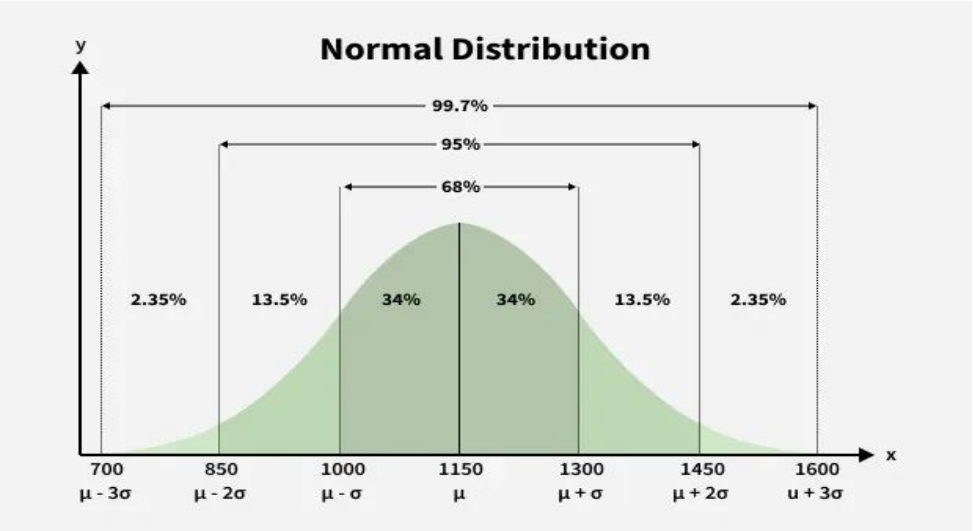
Reasons for choosing this domain:

1. **Relevance:** Exam scores are familiar to students, teachers, and researchers, making the concept easy to relate to.
2. **Predictable behavior:** Most students score around the **average (mean)**, while fewer students score extremely low or extremely high. This produces a **bell-shaped curve**, which is a hallmark of normal distribution.
3. **Visual clarity:** Scores can be easily plotted on a graph to show the **distribution of data** across different ranges.
4. **Application of empirical rule:** Using scores, we can clearly demonstrate the **68-95-99.7 rule**, showing what percentage of students fall within 1, 2, or 3 standard deviations from the mean.

5. **Data analysis insights:** Teachers can analyze student performance, detect **outliers** (very low or very high scores), and take measures to improve learning outcomes.
6. **Educational planning:** Helps in setting grading curves, identifying average performance, and making decisions about class instruction.
7. **Real-life variability:** Scores show natural variation in abilities, study habits, and preparation levels, making them a perfect example to study **variance and standard deviation**.

Example:

- Consider a class of 50 students, where the maximum marks are 100.
- Most students score around **70 marks (mean)**, a few score below **50**, and a few above **90**.
- When plotted on a graph, the data forms a **bell-shaped curve**, showing how the scores are distributed around the mean.
- This curve allows us to **visualize the concentration of scores** near the mean and the spread of other scores.



Explanation of the Normal Distribution Graph with Empirical Rule

Mean (μ):

- The peak of the bell curve represents the **average value** (mean). In your diagram, it's labeled as **μ**.
- Most data points (e.g., students' exam scores) are around this value.

Standard Deviation (σ):

- The distance from the mean to points like **μ+σ** or **μ−σ** shows the **spread of the data**.
- A smaller σ means data is tightly packed around the mean; a larger σ means data is more spread out.

Empirical Rule (68-95-99.7% Rule):

- **68.26% of data** lies within **1 standard deviation** of the mean ($\mu-\sigma$ to $\mu+\sigma$). This is the central part of the curve.
- **95.44% of data** lies within **2 standard deviations** ($\mu-2\sigma$ to $\mu+2\sigma$). Almost all the data is included here.
- **99.72% of data** lies within **3 standard deviations** ($\mu-3\sigma$ to $\mu+3\sigma$). This covers nearly all values.

Tails of the Curve:

- The small percentages at the edges (2.14%, 13.59%) represent **rare extreme values** or outliers.
- For example, in students' exam scores, only a few students will score extremely low ($\mu-3\sigma$) or extremely high ($\mu+3\sigma$).

Symmetry:

- The curve is symmetric around the mean. Left side mirrors the right side.
- This symmetry helps predict probabilities for values above or below the mean.

Example with Students' Scores:

- Mean (μ) = 70, Standard Deviation (σ) = 10
- 68% of students score **60–80**
- 95% of students score **50–90**
- 99.7% of students score **40–100**

This diagram visually shows how **most students score near average**, fewer are below or above, and extreme scores are very rare. It is a **very important tool in statistics** for understanding data distribution and making predictions

Applying the Empirical Rule to Scores

Suppose we have the exam scores of students:

- Mean (μ) = 50
- Standard Deviation (σ) = 10

We want to find out how many students fall into **different score ranges** using the Empirical Rule.

Step 1: Determine ranges using the Empirical Rule

- **68% range ($\mu \pm 1\sigma$):** $50 \pm 10 \rightarrow$ **40 to 60**
- **95% range ($\mu \pm 2\sigma$):** $50 \pm 20 \rightarrow$ **30 to 70**
- **99.7% range ($\mu \pm 3\sigma$):** $50 \pm 30 \rightarrow$ **20 to 80**

Step 2: Example student distribution
Assume 100 students appeared for the exam:

- About **68 students** scored between 40 and 60 (Average Domain).
- About **95 students** scored between 30 and 70 (includes High and Low performers).
- About **99–100 students** scored between 20 and 80, covering almost all students.

This calculation shows how the Empirical Rule predicts the distribution of scores in a simple and visual way.

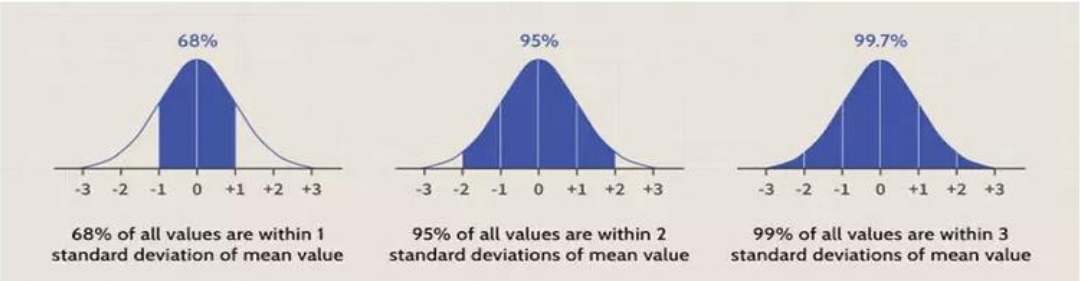
Interpretation of Graph and Data – What the Graph Tells About Student Scores

When we plot the **normal distribution (bell-shaped curve)** for these scores:

- The **highest point** of the curve is at the **mean score (50)**, showing that most students scored around 50.
- The **68% region (40–60)** shows the **Average performers**, who form the majority of students.
- The **95% region (30–70)** includes students who did slightly below or above average.
- The **99.7% region (20–80)** covers nearly all students, including outliers (very low or very high scores).
- Most students are performing near the mean, so the exam was of **average difficulty**.
- Few students are very low or very high scorers, indicating **outliers**.
- This helps teachers understand the **overall performance** and identify who may need extra support or challenge.

Mean, Median, and Standard Deviation – Basic Statistical Terms

In statistics, these three terms are essential for understanding data and its distribution.



Mean (Average):

- The mean is the sum of all data values divided by the number of values.
- It represents the **central value** of the dataset.

Formula:

$$\text{Mean } (\mu) = \frac{\text{Sum of all values}}{\text{Number of values}}$$

- **Example:** For exam scores 60, 70, 80 → Mean = (60+70+80)/3 = 70
- **Interpretation:** Most data points cluster around this value.

Median:

- The median is the **middle value** when data is arranged in order.
- If there is an even number of values, median = average of the two middle numbers.
- **Example:** Scores 60, 70, 80 → Median = 70
  Scores 60, 70, 80, 90 → Median = (70+80)/2 = 75
- **Interpretation:** The median is useful when there are **outliers**, because it is **not affected by extreme values**.

Standard Deviation (σ):

- Standard deviation measures how **spread out the values are** from the mean.
- A small standard deviation means values are close to the mean; a large standard deviation means values are more spread out.

Formula:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

- Where (x_i) = each value, μ = mean, N = number of values
- **Example:** Scores 60, 70, 80 → Standard deviation = 8.16 (approximately)
- **Interpretation:** Helps understand the **variation** in student scores.

Connection to Normal Distribution:

- Mean determines the **center of the bell curve**.
- Standard deviation determines the **width of the bell curve**.
- Median equals the mean in a perfectly normal distribution, confirming **symmetry**.

Conclusion:

From the analysis of students' exam scores using the **normal distribution** and the **Empirical Rule**, we can conclude the following:

1. **Majority of students scored near the mean**, which shows most students performed around the average level.
2. **68% of students fell within one standard deviation** of the mean, representing the Average performers.
3. **95% of students fell within two standard deviations**, including some above-average and below-average students.
4. **99.7% of students fell within three standard deviations**, covering nearly all students and highlighting very few outliers.

Normal Distribution:

- It helps visualize and understand **how data is spread** around the mean.
- Allows teachers and analysts to **identify average, high, and low performers** easily.
- Useful for making **decisions based on data**, such as providing extra help to low performers or challenging high performers.
- The Empirical Rule gives a **quick and simple method** to predict percentages of data within specific ranges.